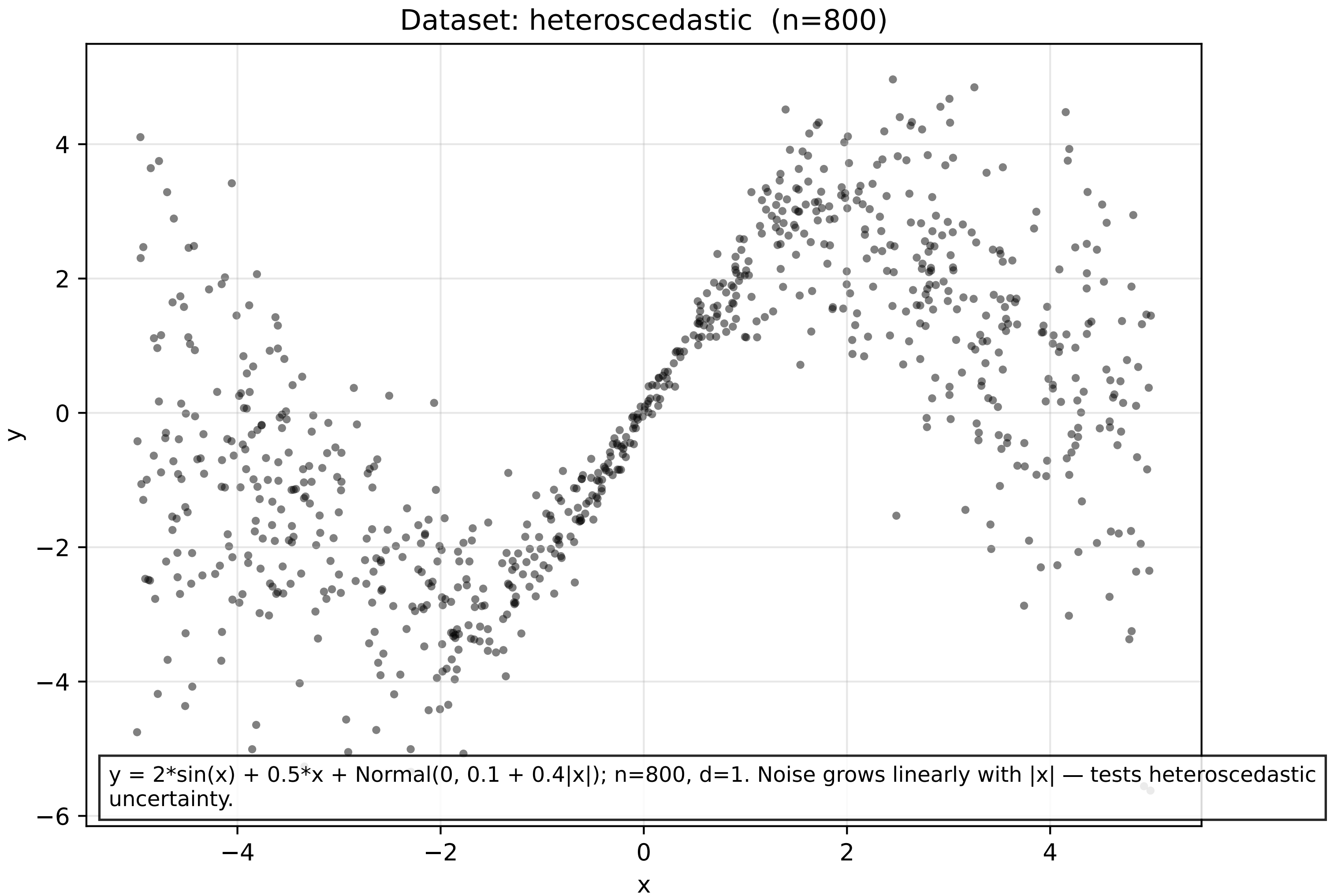
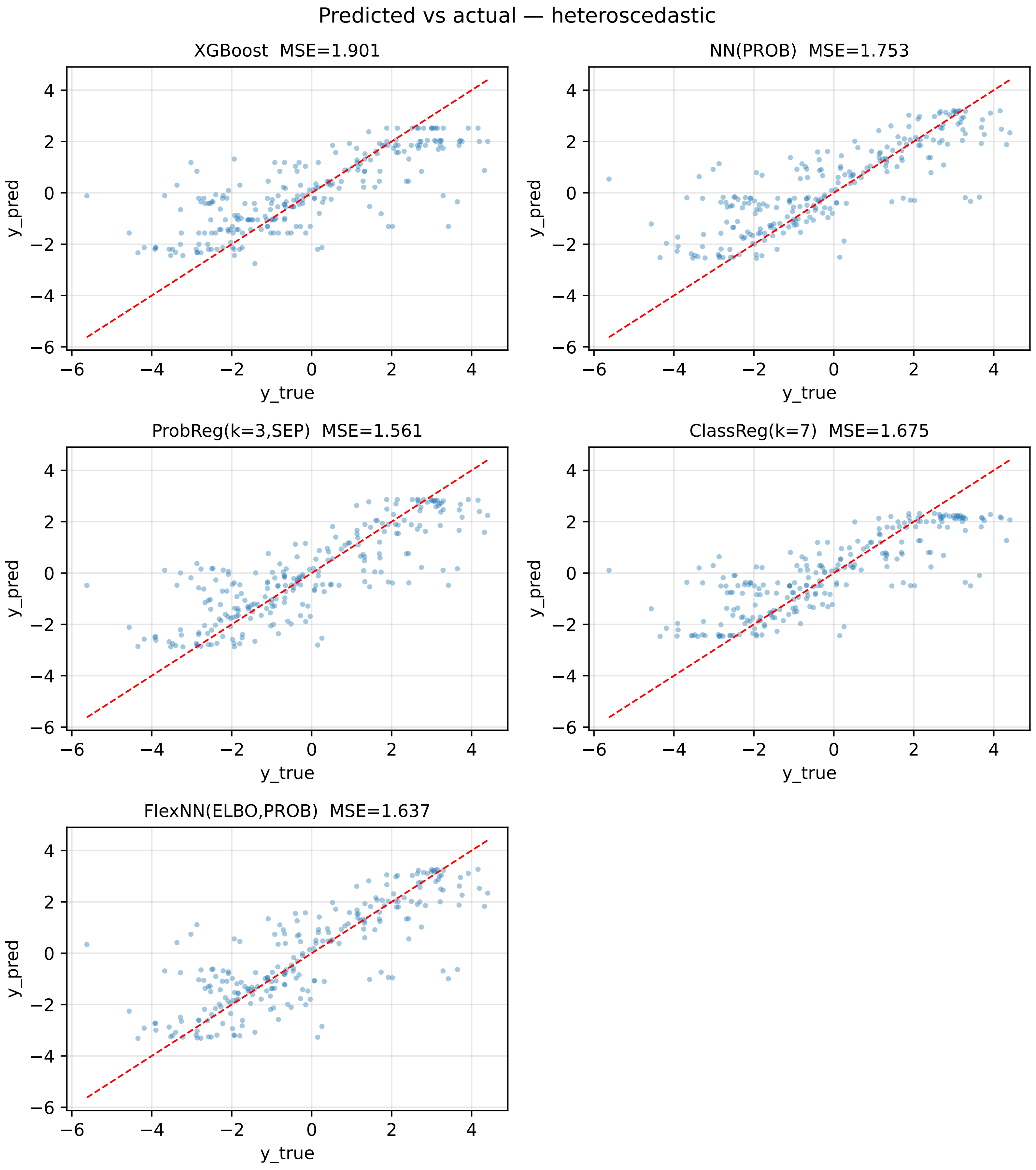
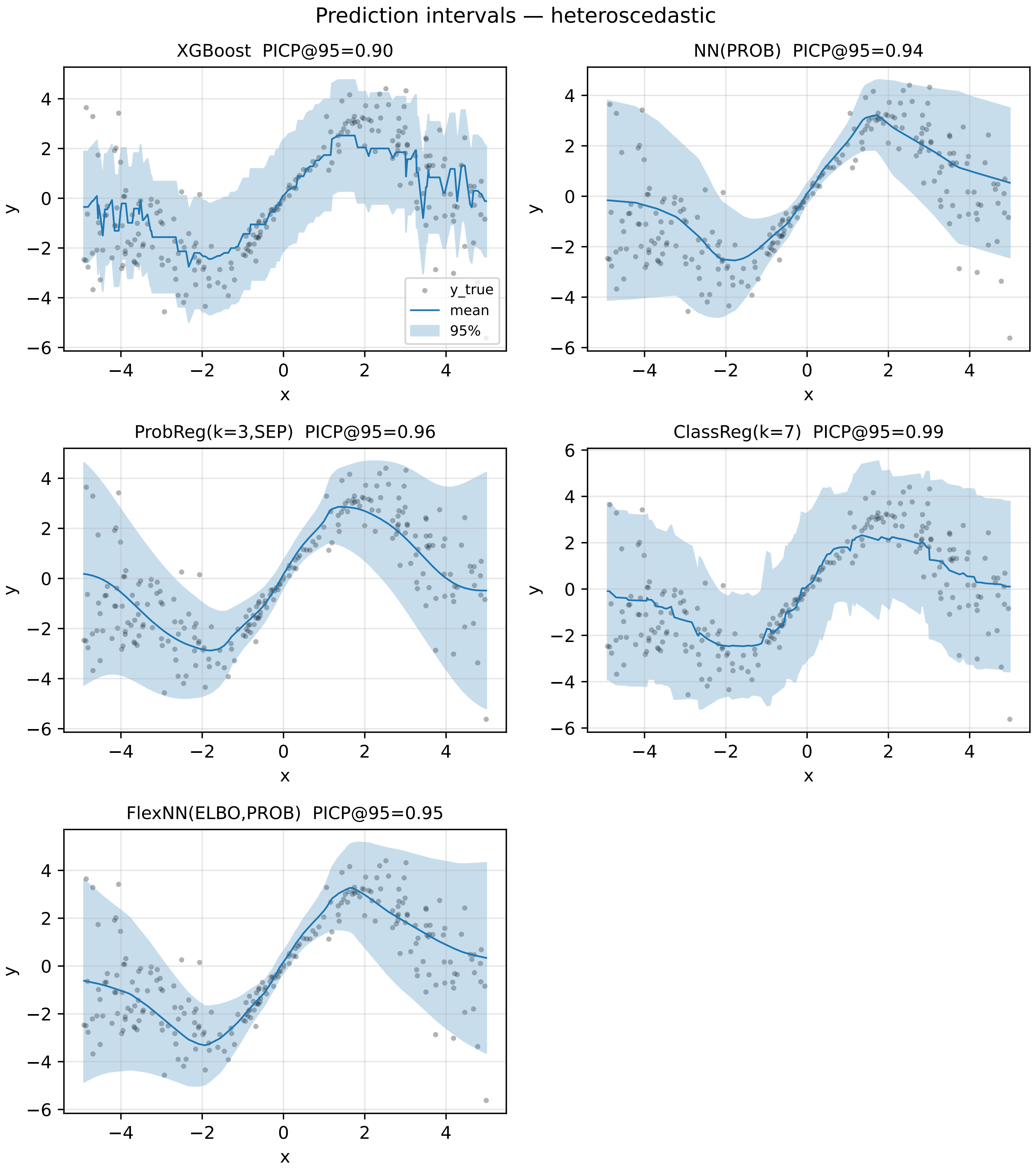
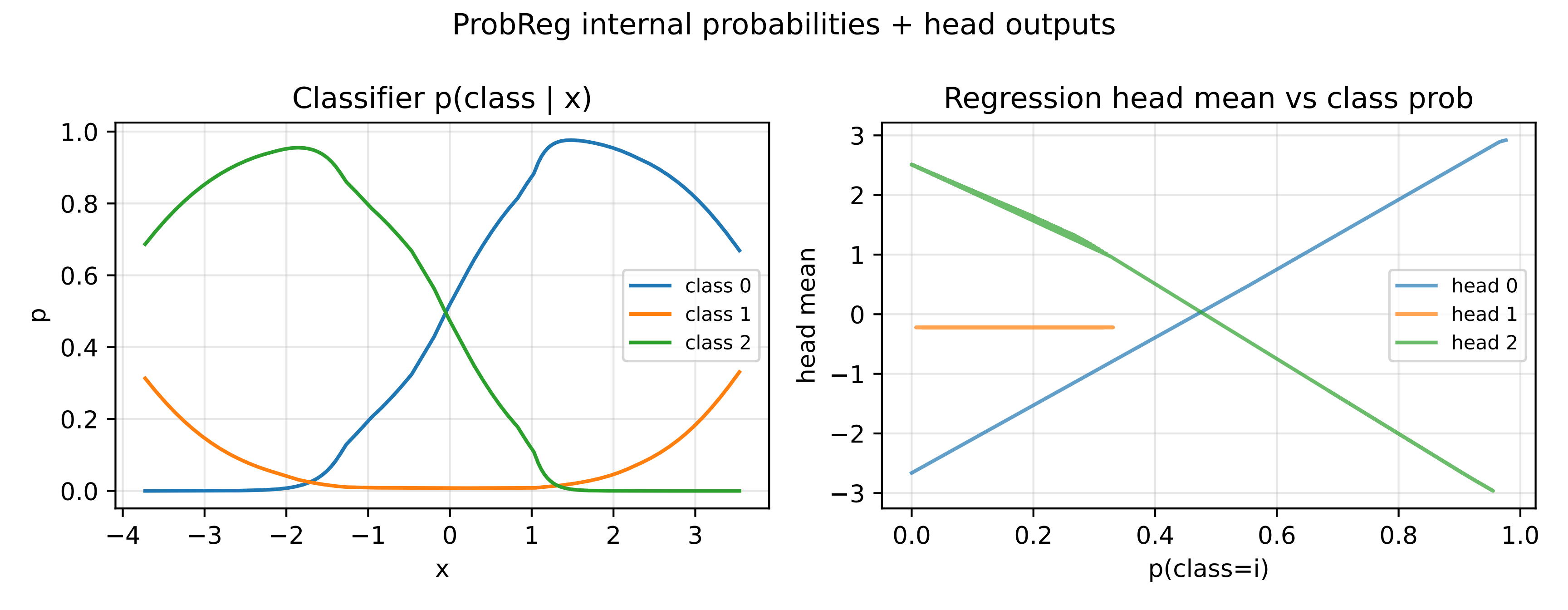

Dataset: heteroscedastic (n=800)
y
x
4
2
0
−2
−4
−6
−4
−2
0
2
4
y = 2*sin(x) + 0.5*x + Normal(0, 0.1 + 0.4|x|); n=800, d=1. Noise grows linearly with |x| — tests heteroscedastic uncertainty.

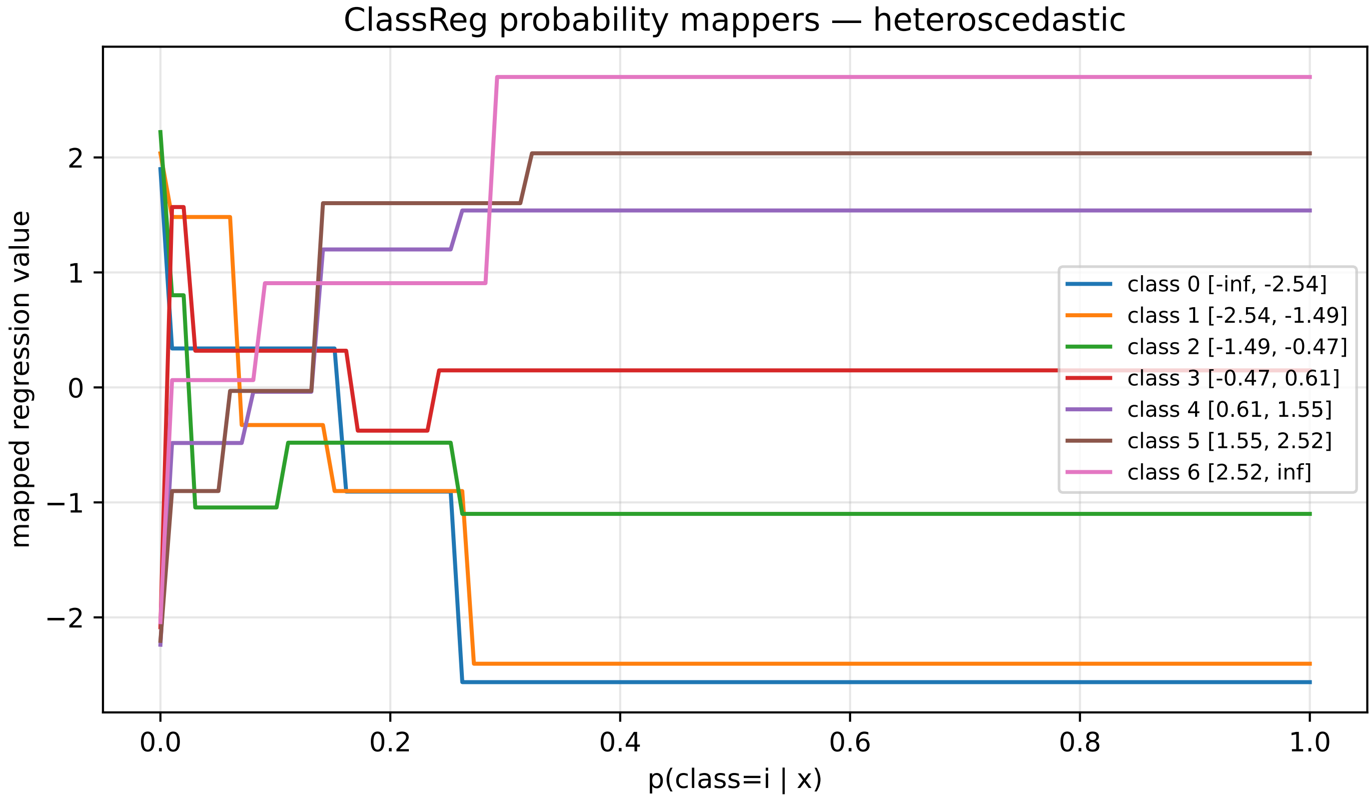
Model comparison

model	mse	nll	crps	picp95	mpiw95	sharpness
XGBoost	1.9012	1.7820	0.7463	0.9042	4.4634	1.1386
NN(PROB)	1.7527	1.4156	0.6615	0.9375	4.4832	1.1437
ProbReg(k=3,	1.5611	1.3856	0.6354	0.9625	4.7598	1.2142
ClassReg(k=7	1.6753	1.6747	0.7096	0.9875	6.3214	1.6126
FlexNN(ELBO,	1.6370	1.4030	0.6393	0.9500	4.6161	1.1776

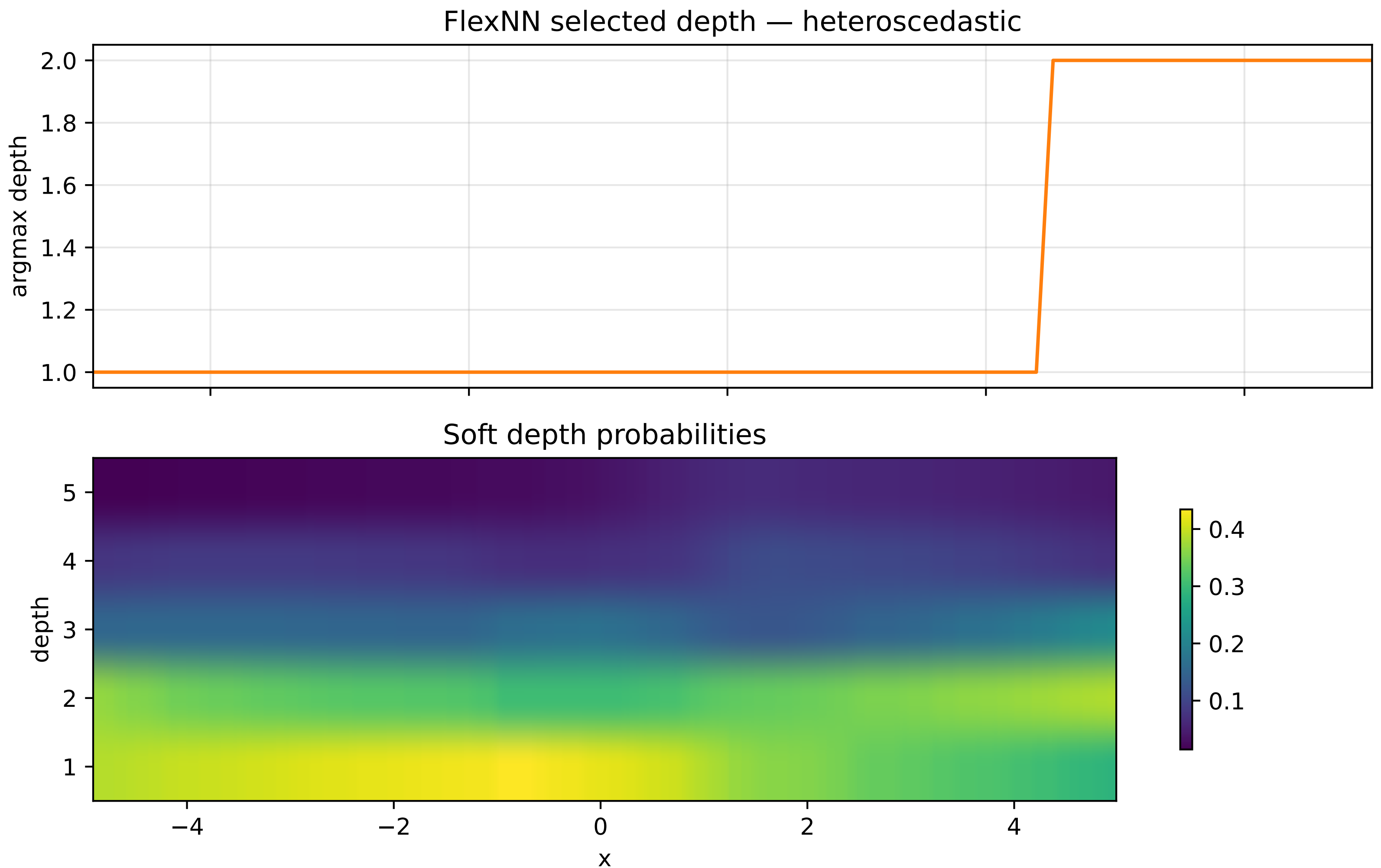

Predicted vs actual — heteroscedastic
XGBoost MSE=1.901
NN(PROB) MSE=1.753
ProbReg(k=3,SEP) MSE=1.561
ClassReg(k=7) MSE=1.675
FlexNN(ELBO,PROB) MSE=1.637
y_pred
y_true


Prediction intervals — heteroscedastic
XGBoost PICP@95=0.90
NN(PROB) PICP@95=0.94
ProbReg(k=3,SEP) PICP@95=0.96
ClassReg(k=7) PICP@95=0.99
FlexNN(ELBO,PROB) PICP@95=0.95
y_true
mean
95%
x
y


ProbReg internal probabilities + head outputs
Classifier p(class | x)
p
x
class 0
class 1
class 2
Regression head mean vs class prob
head mean
p(class=i)
head 0
head 1
head 2



ClassReg probability mappers — heteroscedastic
mapped regression value
p(class=i | x)
2
1
0
−1
−2
0.0
0.2
0.4
0.6
0.8
1.0
class 0 [-inf, -2.54]
class 1 [-2.54, -1.49]
class 2 [-1.49, -0.47]
class 3 [-0.47, 0.61]
class 4 [0.61, 1.55]
class 5 [1.55, 2.52]
class 6 [2.52, inf]



FlexNN selected depth — heteroscedastic
argmax depth
2.0
1.8
1.6
1.4
1.2
1.0
Soft depth probabilities
depth
5
4
3
2
1
−4
−2
0
2
4
x
0.4
0.3
0.2
0.1

```
Ranked MSE for heteroscedastic:
  - ProbReg(k=3,SEP): MSE=1.5611  NLL=1.386  PICP95=0.96
  - FlexNN(ELBO,PROB): MSE=1.6370  NLL=1.403  PICP95=0.95
  - ClassReg(k=7): MSE=1.6753  NLL=1.675  PICP95=0.99
  - NN(PROB): MSE=1.7527  NLL=1.416  PICP95=0.94
  - XGBoost: MSE=1.9012  NLL=1.782  PICP95=0.90

Notes:
Heteroscedastic sin: ProbReg / FlexNN(PROB) should dominate on NLL because the noise scales with
```